

# AutoMLy | Medicaid Policy Copilot

## INAI 6986 — Milestone 1: Data and Model Selection

J.O. • Dr. David M. Simmonds • Evidence through September 28, 2026, 3:42 PM CDT

### Decision and Evidence Boundary

The original September 24 evidence remains historical; later work below is a post-feedback supplement, not evidence that it was available at the original deadline. September 17/18 Muse runs are preserved as historical legacy experiments only; they do not use the new nested split-controlled protocol and are excluded from candidate selection. Gemma training/validation metrics are token-level fit diagnostics, not factual-accuracy comparisons. At the reporting cutoff, Gemma N=1,600 completed at step600/epoch3; checkpoint400 remains its best validation-loss checkpoint. A matched Muse sweep is next. No model has been promoted, and the independent final control/test remain unspent.

## 1. Data Characteristics

The task is a source-cited answer to a public Alabama Medicaid policy or provider-billing question, with a source locator or an abstention when support is absent. It is not an automated eligibility, payment, clinical or legal decision. Each candidate retains source URL, document identity, PDF page, snapshot date and SHA-256 file fingerprint. The target is a response grounded in the frozen source, not a class label.

| Prepared Evidence      | Measured Shape / Consequence                                                                                                              |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Source snapshot        | 138 public PDF files / 2,433 pages (September 14); 136 yielded candidate records.                                                         |
| Ordinary examples      | 1,999 answerable supervised fine-tuning (SFT) records; a separate one-record intentionally unanswerable case was quarantined.             |
| Source mix             | 750 Administrative Code and 1,249 Provider Manual records. Median answer length 18 whitespace-delimited words; 95th percentile 52.        |
| Historical clean split | 1,684 train / 124 validation / 191 historical test records; 113 / 10 / 13 distinct source documents, respectively; zero document overlap. |
| Rights and privacy     | Approved public information only; no patient claims, protected health information or agency-confidential records.                         |

### Preparation and Revised Split

The document-grouped split assigns every page of a source PDF together, preventing page-neighbor leakage. One blank-page safety example was removed from ordinary SFT. The September 16 mixed-generator run was rejected because 326 records came from a retired generator; it is procedural evidence only. September 18 used a different 1,685-row operational render and is not a strict repeat of the clean 1,684-row cohort.

The new experiment nests training subsets of 200, 509, 750 and 1,600 from the same 1,684 clean training rows, covers all 113 training documents even at 200, and holds the same 124 validation rows. Earlier proposals for separate 100/250/200/300-record configuration tests were deferred to protect one final test event. The already exposed historical 191 are retrospective diagnostics, not an untouched final test. The thirty hand-authored control questions remain locked; all their source documents are in the historical test group, none in train or validation. Their question wording and keys are independently authored, but their documents are not independent of the historical test documents.

## 2. Candidates Considered

The declared family is deep learning: local language-model adaptation. Instructor permission covers fine-tuning but not a weakened evaluation standard. The original two Muse runs are one approach plus a sensitivity run, not two architectures. The revised within-family alternatives differ in base architecture, memory demands, tokenizer and deployment feasibility; an eligible training smoke is not a model-quality result.

| Approach                                          | Why Plausible / Trade-Off                                                                                                             | Verified Status                                                                                                                                                                             |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muse Glimmer 30B, rank-16 low-rank adapter        | Legacy four-bit baseline; adapters and historical outputs are preserved for provenance, not reused as matched candidates.             | Sep. 17 clean run and Sep. 18 different-render sensitivity run are historical/non-comparable; excluded from selection. Fresh Muse N=200/509/750/1,600 series is queued after Gemma N=1,600. |
| google/gemma-4-26B-A4B-it, rank-4 adapter         | Different mixture-of-experts architecture and tokenizer; token loss/accuracy cannot rank factual answers across model families.       | N=200, N=509, N=750 and N=1,600 project-data runs complete. N=1,600 stopped at step600/epoch3; best validation checkpoint400. Same fixed validation; no test/control rows used.             |
| nvidia/NVIDIA-Nemotron-3.5-Lightning-30B-A3B-BF16 | Official Megatron Bridge recipes document BF16 LoRA on H100/GB200; local RTX PRO/A6000 backward and checkpoint support is not proven. | Architecture-family documentation reviewed; exact local memory/backward/checkpoint gate remains open; no project-data training.                                                             |
| ornith-ai/Ornith-1.5-35B-A3B                      | Qwen3.5 MoE family; family-level LoRA memory guidance does not prove this exact checkpoint fits or trains locally.                    | Exact backward/checkpoint support and local memory fit remain unverified; no project-data training.                                                                                         |

## 3. Baseline

The current-practice baseline is the student manually finding a source and writing an answer. On thirty simple, hand-authored code/page lookups, all thirty gold answers were source-validated; retrospectively paired session timestamps estimate 42:06 total (about 1:24 each). These are easy, non-blind lookups, not a stopwatch trial and not proof of time savings on policy reasoning. A second technical baseline is untouched Muse on the same frozen generated prompts as the two historical legacy adapters; its human-scored pilot was 4/30. These legacy Muse runs are not candidates in the revised matched experiment. Neither baseline has yet been measured on the proposed harder control.

### Why the Original Control Cannot Decide the Winner

All 30 items ask an answerable page or code lookup. Ten come from Jul26\_34.pdf page 2, six from Jul26\_28.pdf page 2 and five from Apr26\_37.pdf page 2: 21/30 on three pages. This is an honest lookup diagnostic, not a representative policy-support or abstention benchmark. Eight to ten plausible unsupported cases and student-authored exception/date/conflict/multi-rule questions are required before an expanded control is frozen; none has been supplied or scored in this document.

## 4. Evidence and Evaluation

The committed primary outcome is human-adjudicated factual accuracy on answerable questions; unsupported-claim rate, correct locator and abstention are safety-critical secondary outcomes. Existing generated-set results are a small retrospective human pilot. No human-scored comparison on the independent hand-authored control or an expanded reasoning/abstention set has occurred. The table keeps different cohorts and metric types visibly separate rather than pretending they are one same-split ranking.

| Condition / Cohort                                       | Human Factual Accuracy    | Train Fit                                        | Validation Fit                                                                       | Safety / Limits                                                                                                                            |
|----------------------------------------------------------|---------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Untouched Muse; historical generated pilot (n=30)        | 4/30 (13.3%; historical)  | N/A, no tuning                                   | Not measured                                                                         | Retrospective base pilot; one safety response only, insufficient denominator.                                                              |
| Sep. 17 clean Muse; historical legacy pilot (n=30)       | 11/30 (36.7%; historical) | Loss 0.6201 (1,684 rows, 1 epoch)                | Loss 0.5722 (legacy cohort)                                                          | Historical/non-comparable; excluded from selection; 0/1 abstentions; 4/191 truncated.                                                      |
| Sep. 18 Muse sensitivity; historical legacy pilot (n=30) | 9/30 (30.0%; historical)  | Loss 0.6169 (different 1,685-row render)         | Loss 0.5736 (operational cohort)                                                     | Legacy/non-comparable; excluded from selection; not a same-data comparison.                                                                |
| Gemma, N=200 train; same 124 validation rows             | Not scored                | Epoch 3 loss 0.6893; next-token 84.40%           | Epoch 3 loss 0.7044; next-token 83.45%                                               | Train/validation only; 75 steps / 3 epochs.                                                                                                |
| Gemma, N=509 train; same 124 validation rows             | Not scored                | Epoch 5 loss 0.3373; next-token 90.81%           | Best loss 0.6060 at step 128; final loss 0.6636                                      | Best checkpoint 128; run completed step 320; no test/control rows used.                                                                    |
| Gemma, N=750; completed early-stop run                   | Not scored                | Final step 376: loss 0.2192; next-token 93.62%   | Best step 188: loss 0.5829; next-token 85.58%                                        | Completed step 376 after two non-improving validations; no test/control rows used.                                                         |
| Gemma, N=1,600; completed at step 600                    | Not scored                | Step 600/epoch 3: loss 0.2435; next-token 93.05% | Step 600/epoch 3: loss 0.6180; next-token 85.65%; best step 400: loss 0.5750; 85.88% | Early stop at step 600 after two non-improving validations under the 0.005 min-delta rule; best checkpoint 400; no test/control rows used. |

Gemma uses BF16 rank-4 LoRA, learning rate  $1 \times 10^{-4}$ , assistant-response masking and a 1,024-token limit. Next-token accuracy measures prediction of supplied training tokens, not the percentage of policy questions answered correctly. Different tokenizers and training sizes forbid cross-family ranking from token-loss/accuracy columns; no answer-level validation accuracy is reported. N=509's resumed callback lost its early-stop state and continued to step 320; its best saved validation checkpoint remains step 128. N=750 completed early stopping at step 376; checkpoint 188 has the best validation loss. N=1,600 completed at step 600; validation loss was best at step 400 (0.574997). The recorded evaluations at steps 400 and 600 were each less than a 0.005 improvement over the prior best, so the patience-2 callback requested stop. The final summary did not persist a stop-reason field; this classification is reconstructed from its validation history and the runner rule. Use the best validation checkpoint, not the final checkpoint, for within-run analysis.

On the exposed 191-prompt historical generated set, untouched Muse versus the Sep. 17 adapter had answer-token overlap F1 0.2076 versus 0.3973, numeric exact match 6.96% versus 17.39%, and citation-URL presence 0% versus 24.08%. These are legacy engineering diagnostics—not factual accuracy or verified source support—and are not used for selection under the new matched protocol. URL presence mainly reflects output-format behavior. The separate predeclared 30-card human pilot is historical and descriptive; the generated answer key shares the Muse family and human/LLM rating agreement was not blinded. The old promotion check returned pass=false because of truncation and a one-case abstention regression.

## 5. Selection and Justification

No candidate is selected or promoted. The September 17 and September 18 Muse adapters remain historical/non-comparable and are excluded from selection. Gemma N=1,600 completed; next implement the Muse-specific renderer and runner, then run Muse N=200/509/750/1,600 on the same nested train IDs and fixed validation before any matched model-only comparison. Token-level train/validation loss and accuracy support within-setup diagnostics but cannot rank the different tokenizers for factual generalization; the restricted answer-level validation metric is not reported. Nemotron and Ornith remain gated pending exact local memory/backward/checkpoint verification. The independent final test and student-authored control remain unspent; only after a candidate is independently chosen under the authorized criteria may the one-time test compare that winner against untouched base.

## 6. Interpretability and Leakage Check

The practical interpretability check for a language model is whether answers depend on the right exception, effective date, negation and source locator—not merely whether it emits a URL. No paired ablation has been run. Structural checks found zero document overlap, two duplicate-question groups with no cross-split match, one train-validation near-duplicate question candidate at Jaccard  $\geq 0.85$ , and sixteen cross-split duplicate-answer groups affecting fifty-seven records. The group/record counts are reproduced from the canonical file with the project's punctuation-insensitive normalization; Appendix C lists each group. Identical short codes may be coincidental; repeated substantive policy text or template language may transmit information across split boundaries. Neither possibility is a human disposition.

### Open Leakage and Quality Gate

The student must review all 16 answer groups against both questions and source pages, decide which are harmless shared values, boilerplate, or substantive repeated content, and remove/re-group any contaminated held-out item before a clean comparative score. The near-duplicate question pair also needs a source-level ruling. Existing LLM-scored workbook examples are not a human sign-off. A flagged H2011 “Daily” table-column error requires human verification and correction/quarantine before any affected row is reused.

Every existing control item maps by full source URL and SHA-256 to a historical test document; Appendix B makes the per-item result visible. That refutes the specific concern that these thirty questions draw on training documents in the clean lineage, but does not fix their simple wording, source-page concentration, possible generated-test document overlap or absence of unanswerable cases. A revised student-authored control should be frozen separately and excluded from training, development tuning and candidate choice. Plan a sufficiently varied, source-verified answerable stratum and eight to ten plausible-but-unsupported items; score abstention and locator correctness with explicit denominators and a human adjudication key. Do not call the old 1/1 safety probe a 90% estimate.

### Attribution Is a Later Experiment

Open WebUI and ChromaDB retrieval have not been built and deployed for this comparison. No retrieval benefit is claimed. Model-only control scoring can follow a legitimate selection and dataset lock; paired retrieval-on versus retrieval-off attribution waits for the later deployed user interface. This dependency does not excuse unresolved leakage or missing model-only policy-support evidence.

## 7. What Would Change My Mind

A candidate would be rejected or reconsidered if independently scored, source-verified policy questions do not improve answerable factual accuracy by at least 10 percentage points against untouched base, locator correctness is under 95%, unsupported claims reach 2% or more, abstention is under 90% on a meaningful unsupported set, safety routing is under 95%, general/format regression exceeds 3 points, interactive latency fails, or any leakage or rights defect persists. These are project targets, not results. A strong next-token score or a properly formatted URL cannot override human-scored factual and safety failures.

The October 9, 5:00 PM CDT fallback remains a decision gate: if an adapter-loaded or merged model has not produced a complete validated answer, stop adaptation on the critical path and pursue untouched base plus versioned retrieval, reporting adaptation as a negative result. If retrieval later supplies nearly all of the benefit, credit retrieval rather than fine-tuning.

## 8. Repository and Reproducibility

Public course deliverables: [github.com/porespellar/INAI6986\\_project\\_deliverables](https://github.com/porespellar/INAI6986_project_deliverables)

Direct executed historical notebook: [milestone1\\_reproducibility.ipynb](#)

The linked executed notebook rebuilds the historical 1,999-record ordinary dataset, verifies source hashes, reproduces the 1,684/124/191 split, and replays earlier diagnostics. It does not reproduce the new Gemma N=200/509/750/1,600 training, the queued matched Muse series, or any final-control score. The linked public repository hosts the identity-protected J.O. copy of this supplement. The local experiment manifests, run summaries and tested split-rendering scripts are listed in Appendix D; before new metrics are called notebook-reproducible, those artifacts must be incorporated into an executed, portable notebook.

### Evidence Gates Before Comparative Scoring

| Gate                    | Current Disposition / Required Action                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Source membership       | Closed for original 30: 30/30 on historical test documents; zero on clean train/validation. Preserve hash-matched item mapping.     |
| Question validity       | Open: student authors exception, effective-date, conflict and multi-rule items across more sources; original 30 remain lookup-only. |
| Abstention              | Open: add 8-10 independently authored unsupported questions and source-verify the bounded absence of an answer.                     |
| Duplicate/quality flags | Open: human-dispose all 16 groups / 57 records and one near-duplicate question pair; resolve flagged H2011 row.                     |
| Candidate choice        | Open: train/validation token metrics alone cannot rank cross-tokenizer factual generalization; do not peek at protected control.    |
| Final model test        | Not run: lock dataset, recipe, checkpoint, comparator and scorer first; evaluate the frozen candidate and untouched base once.      |

This supplement addresses the feedback with verified repairs and an explicit unfinished-work boundary. It is not a claim that the expanded control, human leakage disposition, final comparison, or a particular course score has been achieved.

## Appendix A — AI Use Disclosure

AI assistance. Hermes Agent (GPT-6 Sol 900k for the initial post-feedback review and GPT-6 Luna 900k for this update) reviewed the professor's feedback, rubric, saved project evidence and document layout; drafted and validated this supplement and its build code. Earlier local Muse generated the original SFT candidates. AI-suggested leakage categories and document prose are not human source adjudication, independently authored control questions, or model outcomes.

Human authorship and accountability. J.O. conceived the project and selected the architecture, source scope, model candidates, metrics, split and guarded workflow; configured local hardware and services, directed agents, diagnosed training and checkpoint failures, and made proceed/stop/quarantine decisions. J.O. wrote and source-verified the original thirty control questions and answer keys. J.O. must author any new control questions, review flagged source evidence and approve final conclusions before submission. The student retains final responsibility for all content and project actions.

Verification. The builder asserts the clean dataset hash, the 30 historical test-document mappings, the 16-group/57-record duplicate count, Gemma N=750 completion at step376, and Gemma N=1,600 completion at step600 with best checkpoint400. DOCX package validation, PDF text readback and page-level visual review are delivery checks, not substitutes for human adjudication.

## Appendix B — Independent Control Source-Split Map

The original source-verified, student-authored set contains thirty answerable lookup questions from ten PDF documents and eleven document/page locations. The three most concentrated pages account for 21/30.

Mapping uses full source URL plus SHA-256 against the canonical f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c dataset; every listed source is in the historical test split. The original questions are preserved verbatim; their source screenshot records remain in the original submission and local evidence directory. The table is a membership audit, not an expanded control or a model score.

| ID     | Original Student-Authored Question                                                                                     | Expected Answer                              | Source PDF / Page                                           | Split |
|--------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------|-------|
| SA-001 | What page number in the table of contents lists Rule 560-X-1-.07, Provider Rights and Responsibilities?                | Page 3.                                      | 9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1                 | test  |
| SA-002 | What page number in the table of contents lists Rule 560-X-28-.02, Summary of State Law Regarding Advanced Directives? | Page 2.                                      | 9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1 | test  |
| SA-003 | What page number in the table of contents lists Rule 560-X-49-.07, Copayment?                                          | Page 2.                                      | 9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1   | test  |
| SA-004 | What page number in the table of contents lists Rule 560-X-50-.03, Recipient Eligibility?                              | Page 3.                                      | 9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1   | test  |
| SA-005 | What Medicaid provider type is assigned to Licensed Social Workers?                                                    | Provider type 11, Crossovers Only.           | Apr26_23.pdf p.1                                            | test  |
| SA-006 | What provider type is assigned to therapy providers?                                                                   | Provider type 17, Therapy.                   | Apr26_37.pdf p.2                                            | test  |
| SA-007 | What provider type is assigned to Remote Patient Monitoring providers?                                                 | Provider type 08, Remote Patient Monitoring. | Jul26_111.pdf p.1                                           | test  |
| SA-008 | What provider type is assigned to physicians?                                                                          | Provider type 31, Physician.                 | Jul26_28.pdf p.2                                            | test  |
| SA-009 | What provider type is assigned to preventive health education providers?                                               | Provider type 55.                            | Jul26_30.pdf p.1                                            | test  |
| SA-010 | What provider type is assigned to psychology providers?                                                                | Provider type 54, Psychologist.              | Jul26_34.pdf p.2                                            | test  |
| SA-011 | What provider type is assigned to Behavioral Health providers?                                                         | Provider type 07, Behavioral Health.         | Jul26_34.pdf p.2                                            | test  |
| SA-012 | What specialty code is listed for Psychology providers?                                                                | Psychology specialty code 112.               | Jul26_34.pdf p.2                                            | test  |
| SA-013 | What specialty code is listed for a Licensed Professional Counselor / LPC?                                             | Specialty code 070                           | Jul26_34.pdf p.2                                            | test  |
| SA-014 | What provider type is assigned to Nurse Practitioners?                                                                 | Provider type 09.                            | Jul26_28.pdf p.2                                            | test  |
| SA-015 | What provider type is assigned to Physician Assistants?                                                                | Provider type 10.                            | Jul26_28.pdf p.2                                            | test  |



**Control Source-Split Map — Continued**

| ID     | Original Student-Authored Question                                                            | Expected Answer                                    | Source PDF / Page | Split |
|--------|-----------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------|-------|
| SA-016 | What specialty code is listed for Occupational Therapy?                                       | Occupational Therapy code 171.                     | Apr26_37.pdf p.2  | test  |
| SA-017 | What specialty code is listed for Physical Therapy?                                           | Physical Therapy code 170.                         | Apr26_37.pdf p.2  | test  |
| SA-018 | What specialty code is listed for Speech Therapy?                                             | Speech Therapy code 173.                           | Apr26_37.pdf p.2  | test  |
| SA-019 | What specialty code is listed for Applied Behavior Analysis Therapy.                          | Applied Behavior Analysis (ABA) Therapy. Code 175. | Apr26_37.pdf p.2  | test  |
| SA-020 | What provider specialties are assigned to Remote Patient Monitoring providers?                | Provider specialties 085 or 086.                   | Jul26_111.pdf p.1 | test  |
| SA-021 | What specialty code is listed for an Associate Licensed Counselor / ALC?                      | Specialty code 071.                                | Jul26_34.pdf p.2  | test  |
| SA-022 | What specialty code is listed for a Licensed Marriage and Family Therapist / LMFT?            | Specialty code 072.                                | Jul26_34.pdf p.2  | test  |
| SA-023 | What specialty code is listed for a Licensed Marriage and Family Therapist Associate / LMFTA? | Specialty code 075.                                | Jul26_34.pdf p.2  | test  |
| SA-024 | What specialty code is listed for a Licensed Master Social Worker / LMSW?                     | Specialty code 073.                                | Jul26_34.pdf p.2  | test  |
| SA-025 | What specialty code is listed for a Licensed Independent Clinical Social Worker / LICSW.      | Specialty code 074.                                | Jul26_34.pdf p.2  | test  |
| SA-026 | What specialty code is listed for a Licensed Psychological Technician / LPT?                  | Specialty code 076.                                | Jul26_34.pdf p.2  | test  |
| SA-027 | What specialty code is listed for Preventive Health Education?                                | Specialty code 183.                                | Jul26_30.pdf p.1  | test  |
| SA-028 | What specialty code is listed for EPSDT in the physician specialty table?                     | EPSDT code 560.                                    | Jul26_28.pdf p.2  | test  |
| SA-029 | What specialty code is listed for Family Practice in the physician specialty table?           | Family Practice code 316.                          | Jul26_28.pdf p.2  | test  |
| SA-030 | What specialty code is listed for Cardiac Surgery in the physician specialty table?           | Cardiac Surgery code 312.                          | Jul26_28.pdf p.2  | test  |

Source concentration (document/page → number of questions): Jul26\_34.pdf p.2: 10; Jul26\_28.pdf p.2: 6; Apr26\_37.pdf p.2: 5; Jul26\_111.pdf p.1: 2; Jul26\_30.pdf p.1: 2; 9.2\_Adm\_Code\_Chap\_1\_General\_3-18-21.pdf p.1: 1; 9.2\_Adm\_Code\_Chap\_28\_Alabama\_Medicaid\_Forms\_7-26-18.pdf p.1: 1; 9.2\_Adm\_Code\_Chap\_49\_Certified\_NP\_Program\_9-23-19.pdf p.1: 1; 9.2\_Adm\_Code\_Chap\_50\_Preventive\_Health\_Ed\_Program.pdf p.1: 1; Apr26\_23.pdf p.1: 1.



## Appendix C — Cross-Split Answer-Overlap Audit

These are reproducible flags, not human rulings. The canonical file yields sixteen punctuation/case-normalized answer groups spanning splits (57 member records). Review the questions, cited source pages and semantic meaning for every group in the human review workbook; a shared page number, repeated boilerplate and repeated policy answer require different remedies. Do not delete records merely because an answer string repeats. Keep the frozen original file and version a corrected split/file after human approval.

| Answer Group (abbreviated)                                          | Members | Splits                | Provisional Risk / Action       |
|---------------------------------------------------------------------|---------|-----------------------|---------------------------------|
| page 2                                                              | 8       | test/train/validation | Needs source-level human ruling |
| this page intentionally left blank                                  | 7       | test/train/validation | Needs source-level human ruling |
| the current procedural terminology cpt and current dental termin... | 6       | train/validation      | Needs source-level human ruling |
| page 3                                                              | 5       | test/train/validation | Needs source-level human ruling |
| 3 90                                                                | 4       | test/train            | Needs source-level human ruling |
| icd 9 codes should be used for claims submitted with dates of se... | 4       | test/train/validation | Needs source-level human ruling |
| 3                                                                   | 3       | test/train            | Needs source-level human ruling |
| 4                                                                   | 3       | train/validation      | Needs source-level human ruling |
| chapter 5                                                           | 3       | train/validation      | Needs source-level human ruling |
| 11                                                                  | 2       | test/train            | Needs source-level human ruling |
| 334 215 7416                                                        | 2       | test/train            | Needs source-level human ruling |
| a hard copy ub 04 claim form must be submitted                      | 2       | train/validation      | Needs source-level human ruling |
| the amount billed shall not exceed the entity s actual acquisiti... | 2       | test/train            | Needs source-level human ruling |
| the current procedural terminology cpt and current dental termin... | 2       | train/validation      | Needs source-level human ruling |
| the ltc gainwell cover sheet from the web portal must accompany ... | 2       | train/validation      | Needs source-level human ruling |
| up to four procedure code modifiers                                 | 2       | test/train            | Needs source-level human ruling |

The one cross-split near-duplicate question pair remains open. The separate quality-review human template and AI example are in artifacts/milestone1/quality\_review/2026-09-24/; no sign-off is inferred from their existence. The original generated-set thirty-card human factual pilot is a different activity.

## Appendix D — Local Evidence Ledger and Sources

| Claim / Requirement                             | Verifiable Artifact                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rubric and instructor's scored feedback         | Assignment 3: Milestone 1 — Data & Model Selection, course rubric; instructor's scored feedback (private course file)                                                                                                                                                                                                                                                                                                                    |
| Historical notebook and original course record  | notebook/milestone1_reproducibility.ipynb; artifacts/milestone1/2026-09-22/{progress_snapshot,evaluation_snapshot}.json; original v2 PDF                                                                                                                                                                                                                                                                                                 |
| Clean dataset and frozen control                | artifacts/milestone1/2026-09-17/approved_candidates.jsonl (SHA-256 f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c); data/curated/student-authored-control-set-2026-09-22.jsonl (SHA-256 a1deb52304d7bce939593a228836debe2359488159e6e3b112e9d48ac7bea55c)                                                                                                                                                              |
| Human pilot and quality flags                   | artifacts/milestone1/human_scoring/2026-09-24/human_generated_sample_scores.json; artifacts/milestone1/quality_review/2026-09-24/INAI6986_1999Record_QualityReview_HumanTemplate.xlsx                                                                                                                                                                                                                                                    |
| Nested train sets and Gemma rendering           | artifacts/model_comparison/2026-09-25/{nested-train-subset-manifest,gemma-render-validation,source-state-inventory}.json; scripts/{prepare_comparison_splits,prepare_gemma_comparison_splits,render_gemma_sft_data}.py                                                                                                                                                                                                                   |
| Gemma N=200, N=509, and N=750 results           | artifacts/model_comparison/2026-09-25/training_results/gemma4-{n200-training-summary,n509-resumed-training-summary,n750-final-training-summary}.json; results.jsonl; run_log.csv                                                                                                                                                                                                                                                         |
| Gemma N=1,600 completed run and best checkpoint | Final step600 checkpoint and completion record are copied locally and SHA-256 verified against amailab1; checkpoint400 remains best validation loss. See training_results/gemma4-n1600-checkpoint600/, gemma4-n1600-checkpoint600.sha256, gemma4-n1600-completion-record/{training_summary,trainer_state,run_config}.json; results.jsonl and run_log.csv. Trainer finished 2026-09-28 15:42 CT, exit0/no OOM; no test/control rows used. |
| Muse methodology change and legacy boundary     | docs/plans/2026-09-28-muse-retraining-methodology-amendment.md; artifacts/model_comparison/2026-09-25/legacy-muse-comparability-note.json; fresh matched Muse series queued after Gemma N=1,600; not started.                                                                                                                                                                                                                            |
| Nemotron/Ornith architecture-family review      | artifacts/model_comparison/2026-09-25/architecture_backward_gate_review_2026-09-28.json; official sources recorded there; exact local backward/memory/checkpoint gates remain open.                                                                                                                                                                                                                                                      |

### Selected Technical Sources

Alabama Medicaid Administrative Code:

[https://medicaid.alabama.gov/content/9.0\\_Resources/9.2\\_Administrative\\_Code.aspx](https://medicaid.alabama.gov/content/9.0_Resources/9.2_Administrative_Code.aspx)

Alabama Medicaid Provider Manuals:

[https://medicaid.alabama.gov/content/Gated/7.6.1G\\_Provider\\_Manuals.aspx](https://medicaid.alabama.gov/content/Gated/7.6.1G_Provider_Manuals.aspx)

LoRA method: Hu et al., <https://arxiv.org/abs/2106.09685>. Candidate model identities and exact pinned revisions are recorded in artifacts/model\_comparison/2026-09-25/source-state-inventory.json; no third-party performance claim is inferred from a model card.